

NORAN ESSAM

PMP® | Mechanical Engineer – HVAC & AI Automation Enthusiast– Manufacturing Industry

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PROFILE SUMMARY

PMP®-certified Mechanical Engineer and Project Leader with 4+ years of experience in HVAC manufacturing and engineering operations. Proven track record in leading cross-functional teams, optimizing project workflows, developing KPI dashboards, and supporting on-time delivery through data-driven decision-making. Strong expertise in project coordination, risk mitigation, resource planning, and stakeholder communication, with hands-on experience in Power BI and advanced Excel. Passionate about combining engineering expertise with AI technologies to modernize traditional workflows.

EDUCATION & CERTIFICATIONS

B.Sc. Mechanical Engineering, Alexandria University, 2021
Project Management Professional PMP® – PMI, Dec 2025-Dec2028, Cert. #4252282

PROFESSIONAL EXPERIENCE

AL SAMMAN For Air Distribution | Senior Technical Office & Estimation Engineer April 2026 – Present
Ductwork & Accessories Supplier Factory | project Supplier

- **Cost Estimation:** Preparing Bill of Material (BOM) calculation and costing processes, including raw material analysis, manufacturing cost estimation, labor Costs and Full pricing calculations for HVAC products.
- **Process Improvement:** Developed an AI-powered quotation automation system that transformed HVAC BOQ processing into a semi-automated workflow, significantly reducing quotation turnaround time while improving estimation consistency and operational efficiency.  Portfolio & Demo: [AI Quotation Automation Portfolio](#)
- **Technical Coordination:** Managed the end-to-end preparation of technical submittals, prequalification documents, and compliance records for complex HVAC projects.
- **Data Management:** Supported engineering teams with precise data collection and reporting to facilitate accurate project documentation.

Rabie Al-Takyeef Factory | CSD Operations Team Leader Aug 2024 – Feb 2026
Manufacturing company specializing in HVAC systems

- **Team Leadership:** Lead and mentor a cross-functional team of **18 engineers and data specialists** , ensuring alignment with project milestones and quality standards.
- **Problem Solving:** Collaborated with internal stakeholders to resolve **20+ critical technical issues** , directly reduces project delays and ensuring schedule adherence.
- **Process Optimization:** Spearhead data-driven workflows using **Root Cause Analysis (RCA)** to standardize operations, resulting in significant Improvement in team efficiency and data quality.
- **Data Analysis & Reporting:** Developed and deployed **5+ KPI dashboards in Power BI** to track project progress, workload distribution, and performance, providing actionable insights to senior management for strategic decision-making.
- **Mentoring & Training:** Designed and delivered HVAC training curriculum; mentored 15+ junior engineers & data entry specialists.
- **Resource Management:** Conducted strategic workforce planning to forecast project needs, leading the successful recruitment and training of **10 new team members** to scale production capacity.
- **Risk Management:** Acted as the primary escalation point for project risks, successfully mitigating **3 major schedule overruns** through proactive intervention and resource reallocation.
- **Cross-Functional Collaboration:** Partnered with manufacturing and engineering departments to validate technical considerations and optimize resource allocation for project scopes.

- **Technical Coordination:** Managed the end-to-end preparation of technical submittals, prequalification documents, and compliance records for complex HVAC projects.
- **Technical Office:** Preparing technical offers & quotations, technical submittals, pre-qualifications, product selections, and providing customer support on technical needs for air outlets, dampers, VAVs, louvers, ductwork, and sound attenuators in compliance with SMACNA, ASHRAE, AHRI, AMCA, and UL standards.
- **Acoustic Engineering Analysis:** Conducted Full acoustic path analysis per AHRI 885 for commercial and institutional projects; achieved NC targets to ensure all deliverables meet rigorous industry standards.

PRODUCTS & MANUFACTURING EXPERIENCE

Extensive experience across the full HVAC air-distribution product range:

- Strong knowledge of sheet metal manufacturing processes including shearing, bending, folding, and spinning operations for HVAC products and ductwork fabrication.
- Ductwork systems design, fabrication, and standards compliance in accordance with SMACNA requirements.
- Supply Air Registers & Grilles — single/double deflection, Bar Grills, ceiling & sidewall types per ASHRAE 70
- Ceiling Diffusers — square, round, slot; ASHRAE 70 certified performance selection
- Volume Control Dampers — motorized and manual; actuator torque sizing per AMCA 500-D
- Fire / Fire & Smoke Dampers — per Fire & Safety Codes & Regulations UL 555 & UL 555s
- Louvers & Weather Protection — free area calculations, water penetration ratings as per AMCA 500-L
- VAV Boxes — single-duct, Air Terminals; capacity, noise, pressure selection - AHRI
- Sound Attenuators — rectangular/circular silencers; insertion loss analysis.

Standards Applied:

- AHRI 885 — occupied space sound estimation
- ASHRAE 68.1 — measurement of sound attenuation in silencers
- ASHRAE 70 — terminal air device acoustic performance
- ASHRAE Handbook Applications Ch. 48 — Noise & Vibration Control

SKILLS

- **Technical Skills:** AutoCAD, SolidWorks, HVAC Operations Principles, MS Excel (Advanced: Formulas, Pivot Tables, Charts), Power BI, Data Analysis (Data Cleaning, Visualization, Trend Analysis)
- **Project Management & Leadership:** Project Management Professional (PMP), Project Planning & Scheduling, Scope Coordination, Team Leadership & Mentoring, KPI & Progress Reporting, Process Improvement, Stakeholder Communication, Risk & Issue Management, Resource Coordination, PMBOK-Aligned Project Management Methodologies
- **Soft Skills:** Excellent Communication, Customer Relations, Multitasking, Decision-Making, Problem-Solving
- **Languages:** English (Excellent), Arabic (Native)

ACADEMIC PROJECTS & LEADERSHIP

Formula SAE Italy 2019 - University Racing Team 2019

- Authored the comprehensive marketing strategy and business plan for the team's competition entry, contributing to the overall project score.
- Conducted market research and cost analysis to define the vehicle's target market and projected production cost.
- Collaborated with engineering and design teams to ensure the business case is aligned with technical feasibility and competition rules.